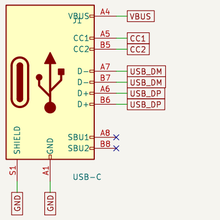
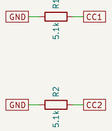
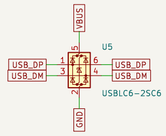
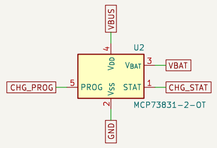
USB-C, ESD protection

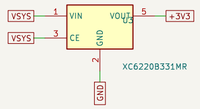


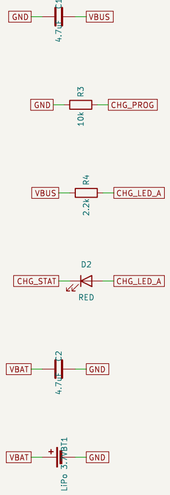


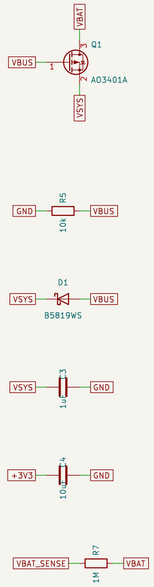


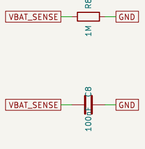
Charger, battery, power path, 3.3 V regulator



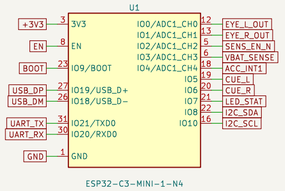


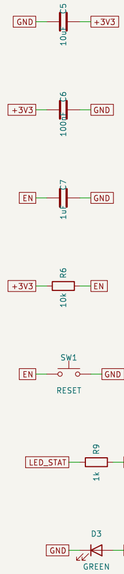






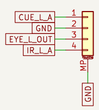
ESP32-C3 module, buttons, status LED, debug pads

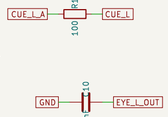


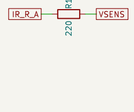


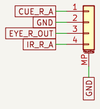


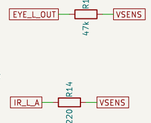
Eye sensor connectors and sensor power switch

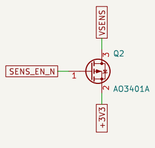




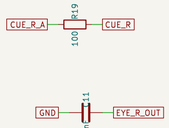


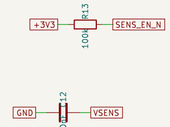


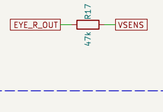




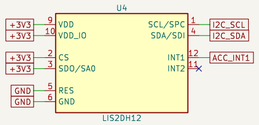


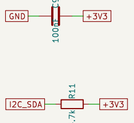


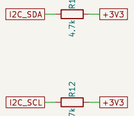


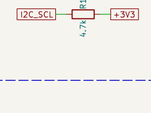


Accelerometer











Charge current = 1000 V / R3 = 100 mA. Use a cell of 100 mAh or more with its own protection circuit.
Q1 and D1 share the load: on USB power the system runs from VBUS and the battery only charges.

Q2 switches power to both eye sensors. SENS_EN_N low = sensors on.
Strapping pins: GPIO2 (SENS_EN_N, pulled up by R13), GPIO8 (SDA, pulled up), GPIO9 (BOOT pad, low at reset = USB download).
Accelerometer I2C address 0x19 (SA0 high).
Eye connector pins: 1 cue LED anode, 2 GND, 3 phototransistor output, 4 IR LED anode.

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Title: DreamCatcher main board

Size: A2	Date:	**Rev: 1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1